

Models Evaluated

Model	Architecture	Parameters
Whisper Large V3	Transformer Encoder-Decoder	1.55B
IndicConformer-600M Multilingual	Conformer-CTC	600M
IndicWav2Vec Hindi	Wav2Vec 2.0	300M
IndicWav2Vec Bengali	Wav2Vec 2.0	300M
Whisper Medium Telugu	Transformer Encoder-Decoder	769M
Vaani FastConformer Telugu	FastConformer-CTC	120M
Vaani FastConformer Hindi	FastConformer-CTC	120M

Test Dataset

Dataset	Languages	Description	Open Source	Approximate Size
Kathbath	12+ Indian languages	Crowdsourced read speech dataset for ASR research	Yes	~7k hrs
IndicVoices	22 Indian languages	Large-scale crowdsourced speech corpus covering diverse accents and domains	Yes	~17k hrs

Evaluation metrics

1)Word Error Rate

Measures the percentage of incorrectly recognized words.

$$\text{WER} = (S + D + I) / N$$

Where:

- **S** = Word Substitutions
- **D** = Word Deletions
- **I** = Word Insertions
- **N** = Total words in the reference transcript

Lower WER indicates better ASR performance. WER is the most widely used metric for evaluating ASR systems

2)Character Error Rate

Measures errors at the character level instead of the word level.

$$\text{CER} = (S + D + I) / N$$

Where:

- **S** = Character substitutions
- **D** = Character deletions
- **I** = Character insertions
- **N** = Total characters in the reference transcript

Lower CER indicates better performance. CER is especially useful for languages with rich morphology when evaluating multilingual ASR systems.

Per-Language WER/CER Results

Table 5 presents the WER and CER of the top-3 models for each of the five languages:

Language	Model	WER(%)	CER(%)	Test Set	Test Set Size	Rank
Hindi	Vaani-FastConformer-Hindi	15.11	7.09	IndicVoices-Hindi	5530	1
	IndicConformer-600m-multilingual	15.3(RNNT) 16.6(CTC)	7.2(RNNT) 7.3(CTC)	IndicVoices-Hindi	5530	2
	IndicWav2Vec-Hindi	38.6	22.5	IndicVoices-Hindi	5530	3
Bengali	IndicConformer-600m-multilingual	13.49(RNNT) 14.23(CTC)	5.36(RNNT) 5.41(CTC)	IndicVoices-Bengali	3906	1
	IndicWav2Vec-Bengali	46.7	21.2	IndicVoices-Bengali	3906	2
Marathi	IndicConformer-600m-multilingual	14.9(RNNT) 16(CTC)	5.61(RNNT) 5.69(CTC)	IndicVoices-Marathi	3552	1
Telugu	IndicConformer-600m-multilingual	26.28(RNNT) 27.86(CTC)	8.42(RNNT) 8.53(CTC)	IndicVoices-Telugu	3295	1
	Vaani-FastConformer-Telugu	27.31	8.77	IndicVoices-Telugu	3295	2
	ARTPARK-IISc/whisper-medium-vaani-telugu	61.69	25.69	IndicVoices-Telugu	3295	3
Tamil	IndicConformer-600m-multilingual	30.7(RNNT) 34.7(CTC)	9.1(RNNT) 10.1(CTC)	IndicVoices-Telugu	5276	1

Challenges & Solutions

1. Inference Speed vs. Resource Usage

Challenge: Using a streaming dataset requires sequential data loading, which slows down inference. In contrast, non-streaming datasets must be downloaded from Hugging Face, increasing disk space requirements but enabling faster batch inference.

Solution: Used non-streaming datasets when faster batch inference was needed and sufficient storage was available. Used streaming datasets in storage-constrained environments where minimizing disk usage was more important than inference speed.

2. Invalid Dataset Samples

Challenge: Some dataset entries contained missing audio, incorrect transcripts, or corrupted metadata.

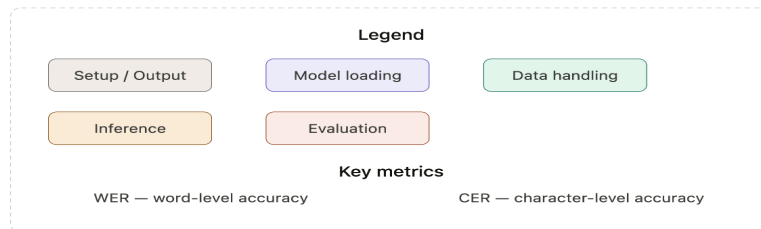
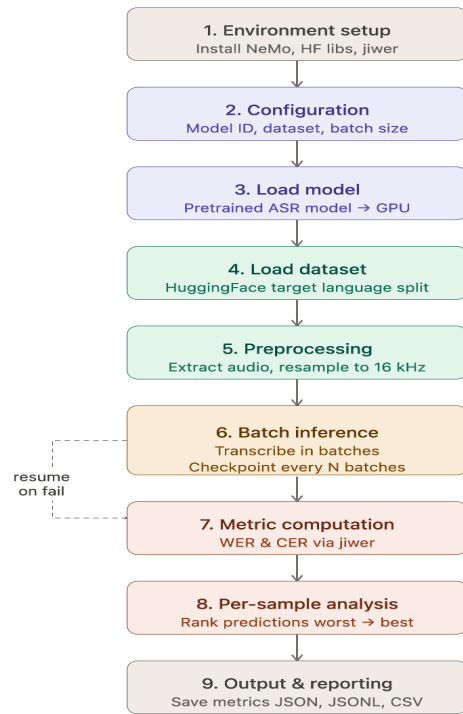
Solution: Implemented sample-level validation and error handling to skip problematic entries without interrupting evaluation.

3. Pipeline Interruptions

Challenge: Long-running inference jobs can be interrupted by session timeouts or runtime failures.

Solution: Implemented checkpointing to periodically save progress and resume evaluation from the last completed batch.

Flowchart of Evaluation



Conclusion and Next Steps

Key Observations:

- **IndicConformer-600M Multilingual (RNNT)** achieved the best performance across all evaluated languages and is the strongest overall multilingual ASR model.
- **RNNT decoding** consistently outperformed **CTC decoding** in both WER and CER across all languages.
- **Vaani-FastConformer-Hindi** slightly outperformed IndicConformer on Hindi, making it the best choice for Hindi-only applications.
- **Telugu** showed close competition between IndicConformer and Vaani-FastConformer-Telugu, with both delivering similar performance.
- **IndicWav2Vec** models performed significantly worse than Conformer-based models and are less suitable for production use.
- **Bengali** achieved the lowest WER, indicating strong model performance and dataset coverage.
- **Tamil** was the most challenging language, with the highest error rates among the evaluated languages.

Overall, **IndicConformer-600M Multilingual with RNNT decoding** is the recommended baseline for multilingual Indian ASR systems.

Next Steps:

- Evaluate the remaining ARTPARK-IISC models using all 5 languages.
- Fine tune ASR models using custom datasets.